

Related U.S. Application Data

continuation of application No. PCT/CN2020/
070110, filed on Jan. 2, 2020.

(56)

References Cited

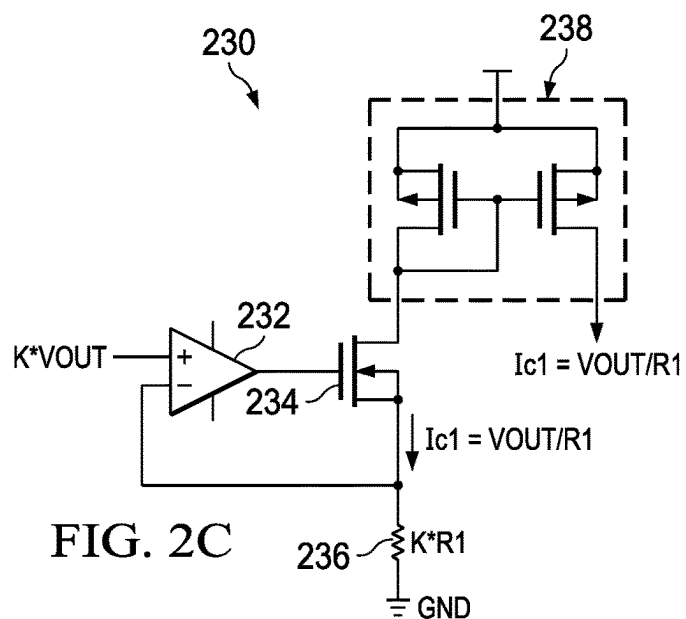
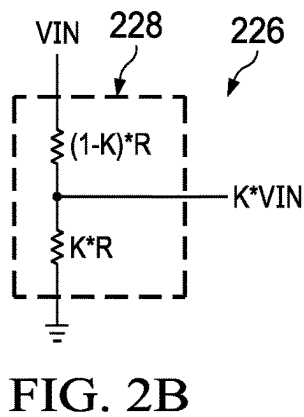
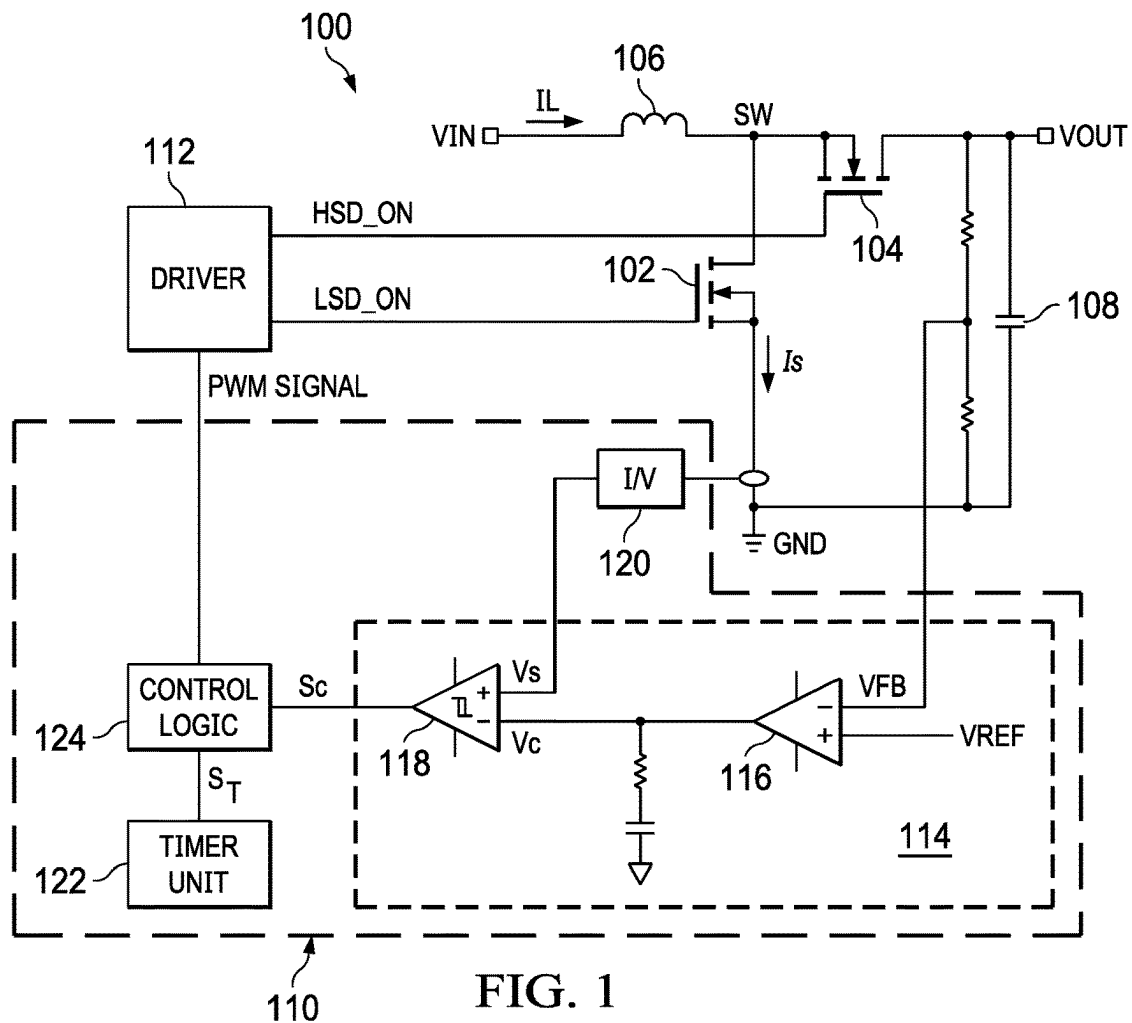
U.S. PATENT DOCUMENTS

10,992,231	B1	4/2021	Yang	
2005/0134247	A1	6/2005	Lipcei	
2008/0088284	A1	4/2008	Weng	
2008/0088292	A1 *	4/2008	Stoichita	H02M 3/156 323/285
2009/0140708	A1	6/2009	Tateishi	
2013/0009557	A1	1/2013	Szczeszynski	
2017/0302180	A1	10/2017	Villar Piqué	

OTHER PUBLICATIONS

English machine translation for CN101540548A, 20 pages.
English machine translation for CN102714462A, 52 ages.
English machine translation for CN103378740A, 22 pages.
First Office Action in Chinese counterpart Application No.
202080091587.5, dated May 25, 2023.

* cited by examiner



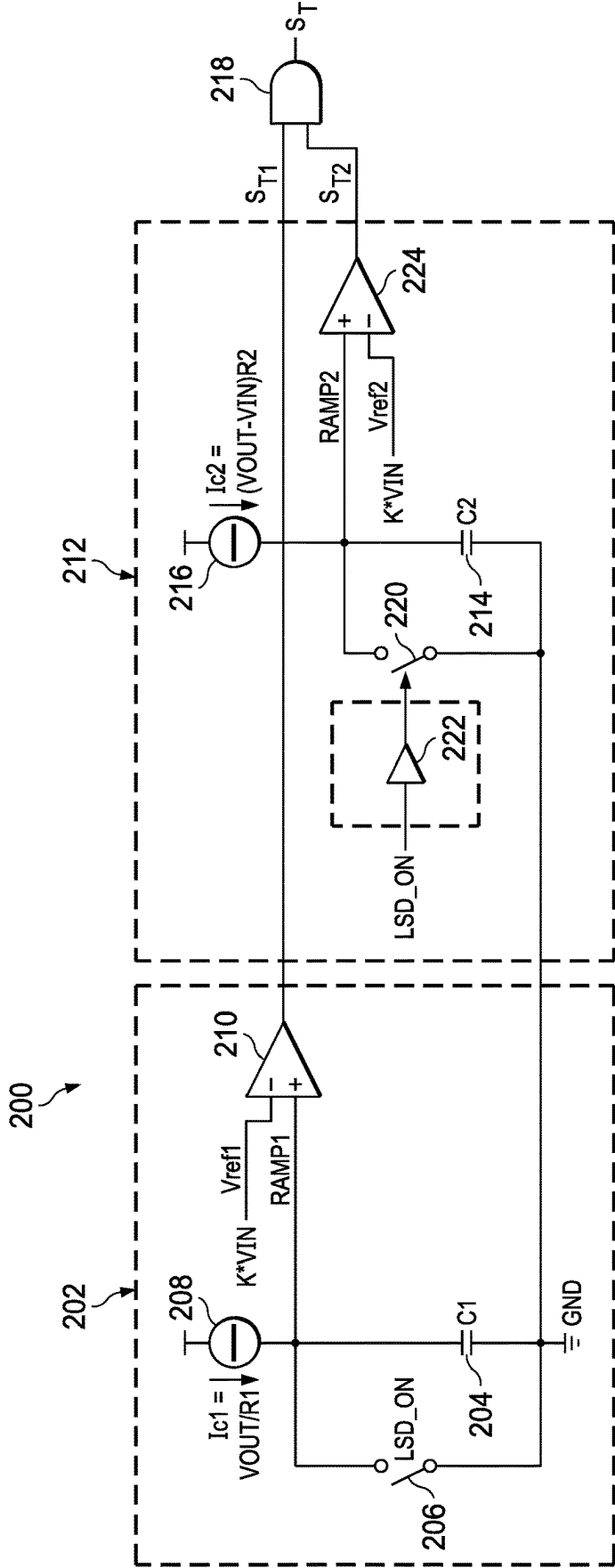


FIG. 2A

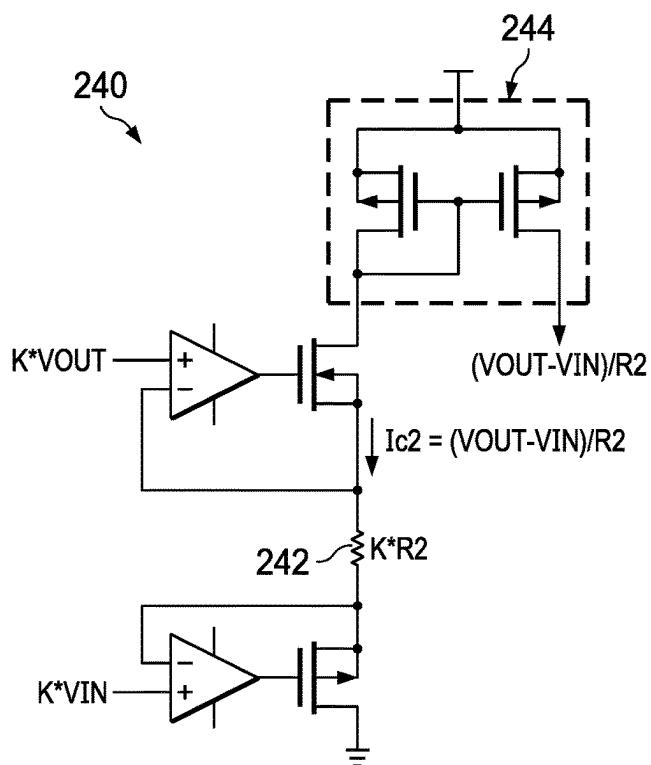


FIG. 2D

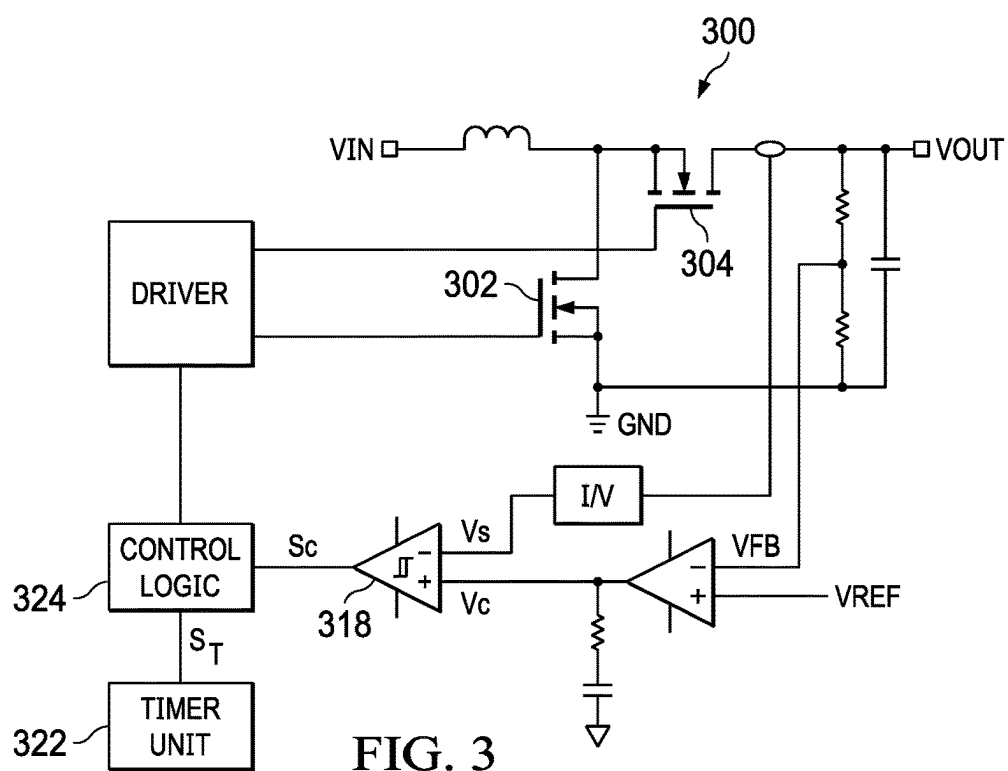


FIG. 3

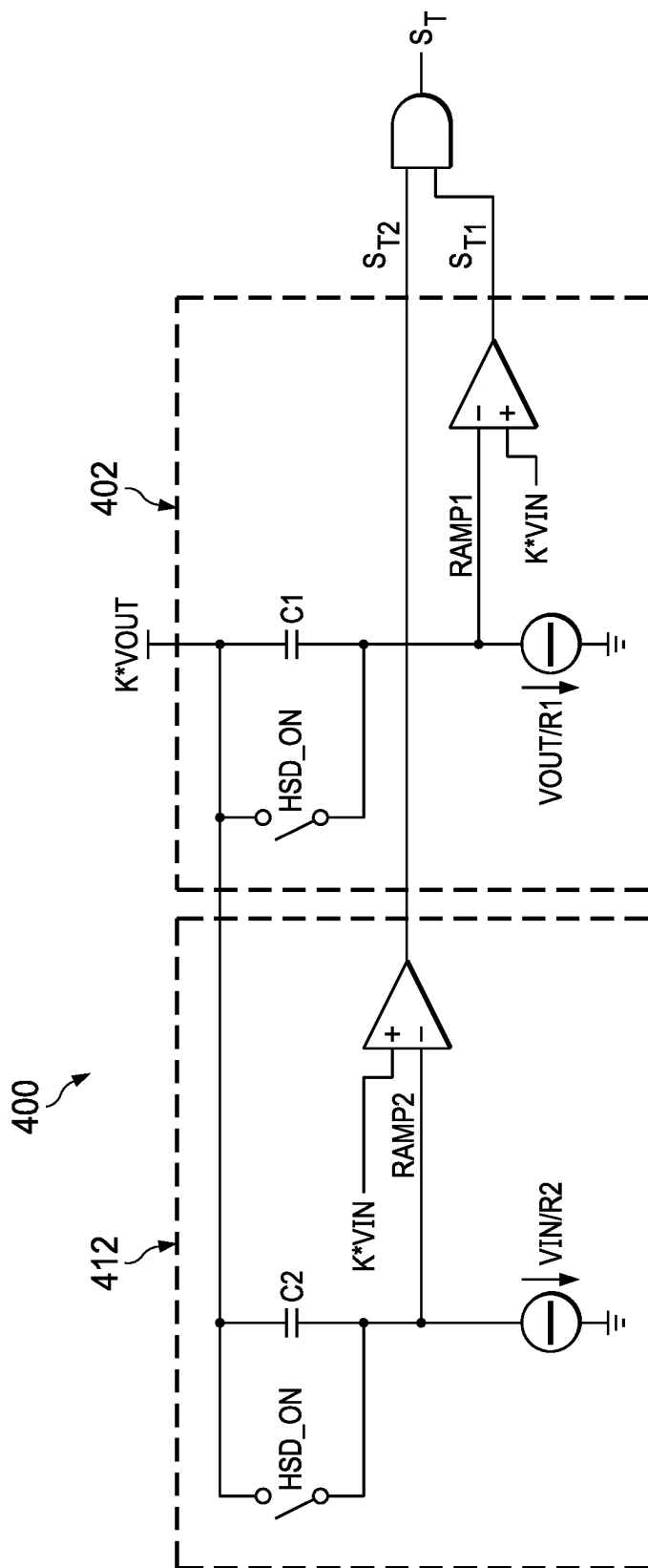


FIG. 4

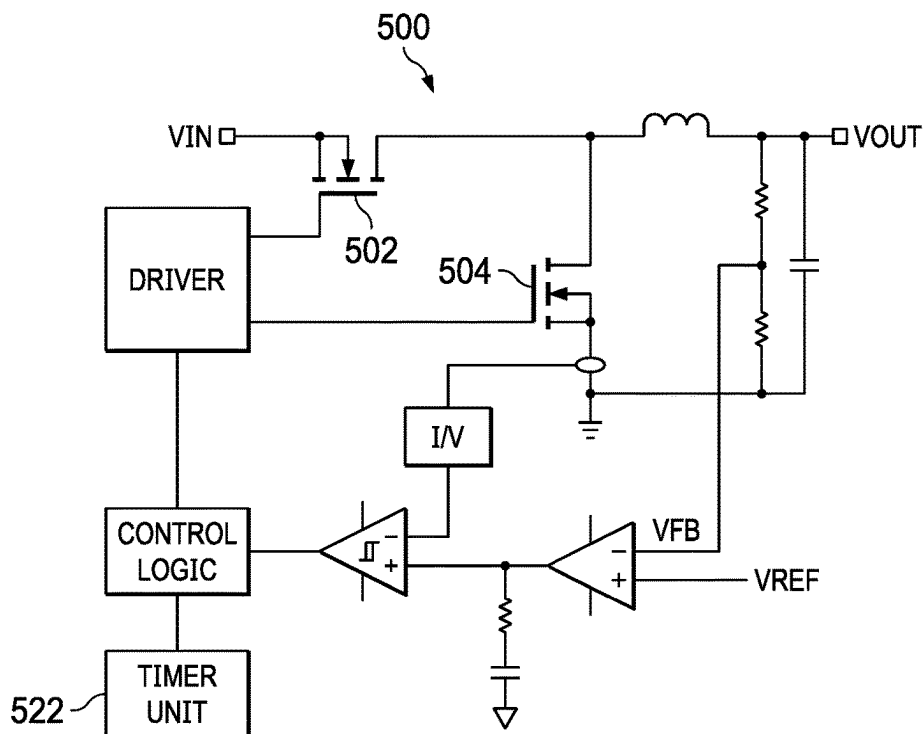


FIG. 5

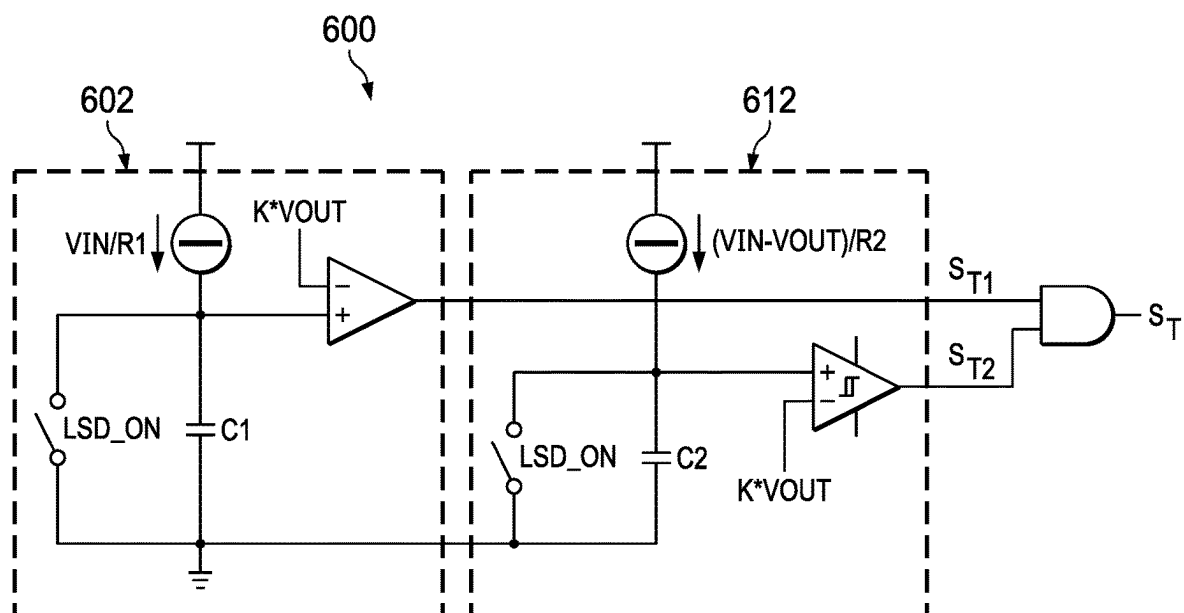


FIG. 6

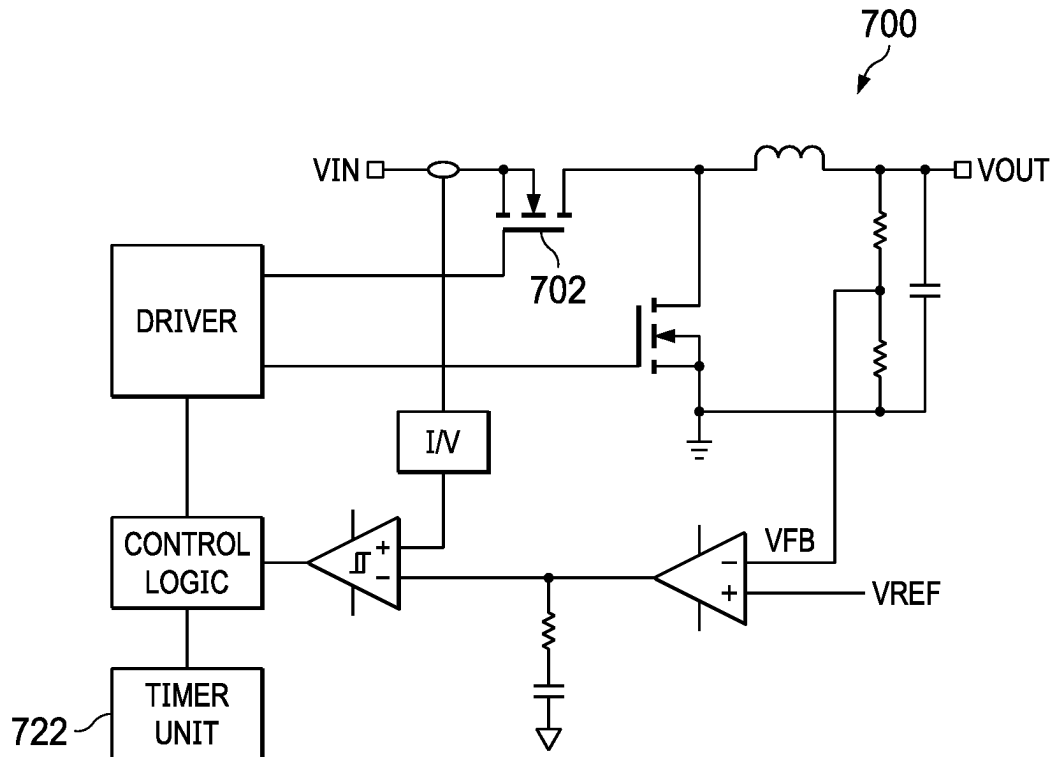


FIG. 7

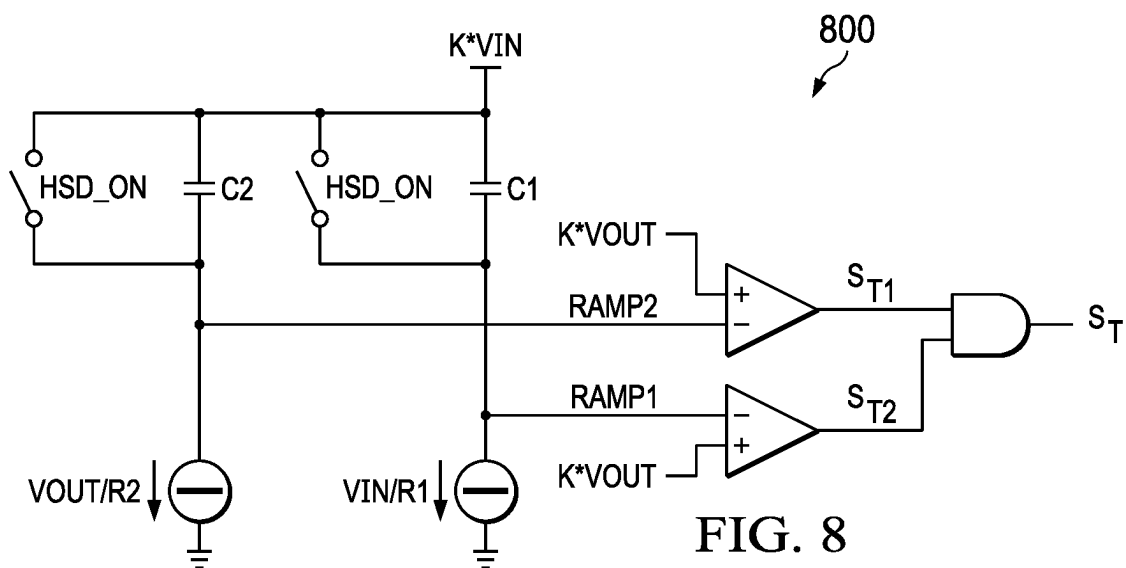
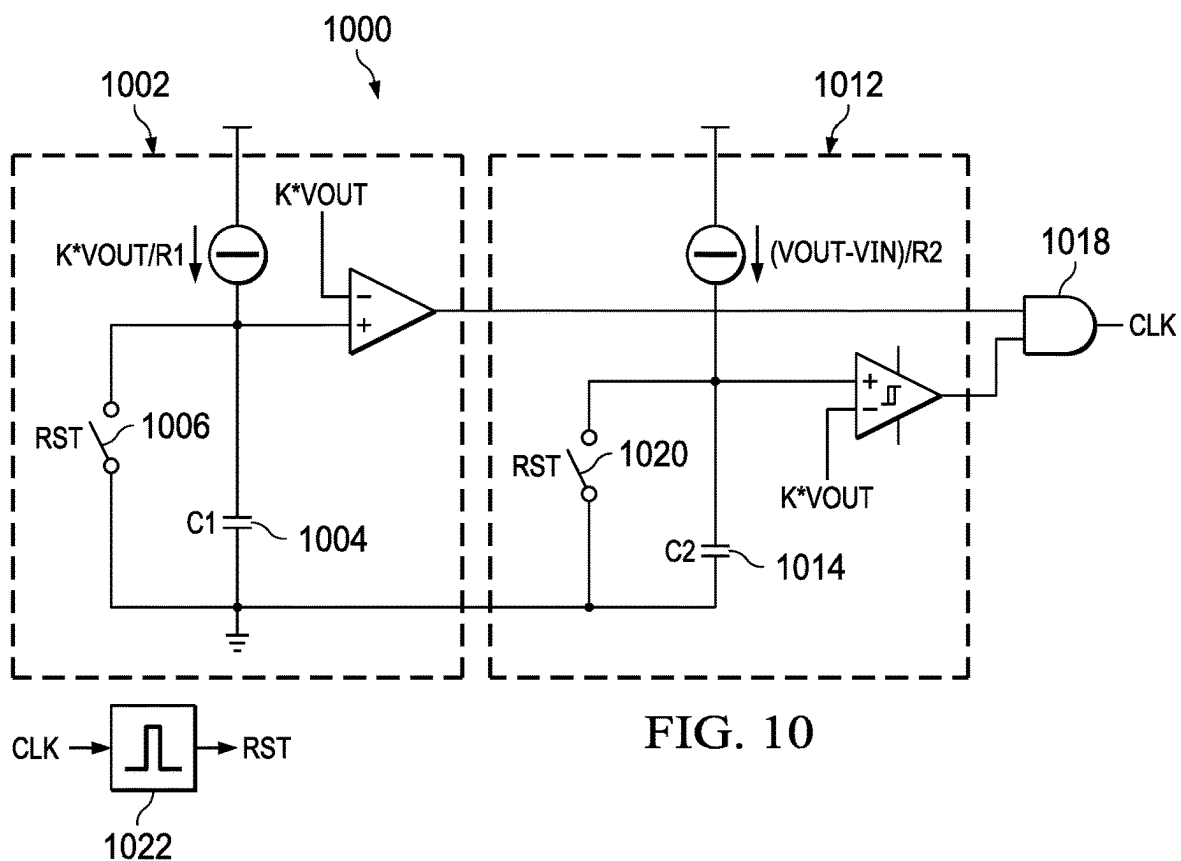
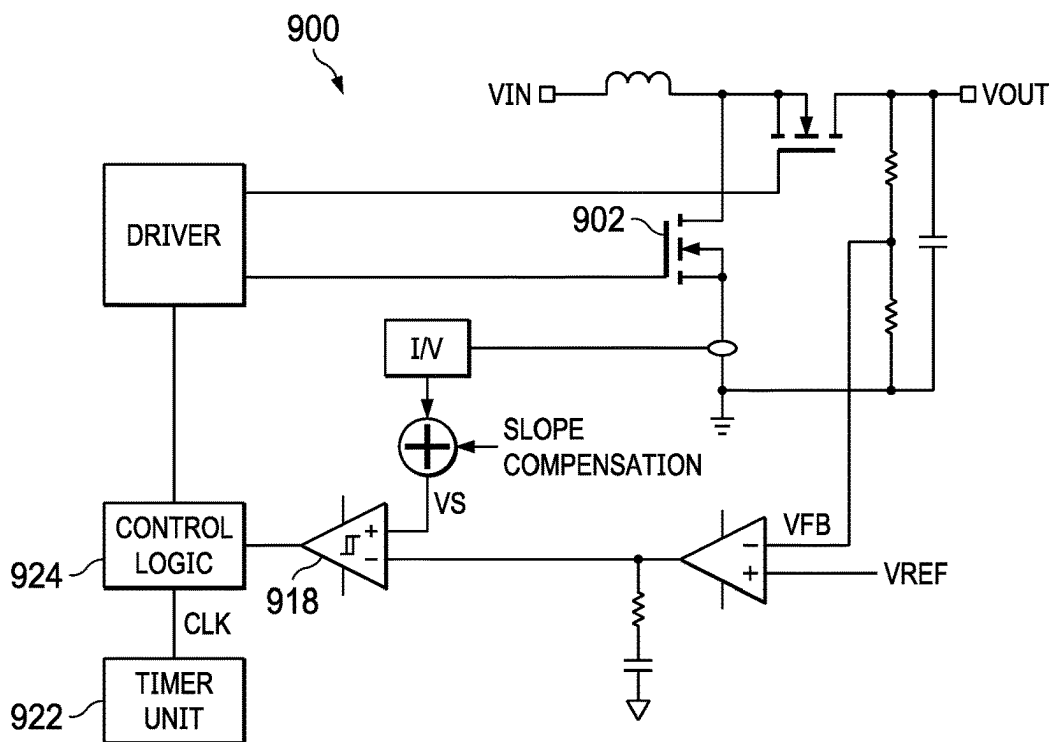


FIG. 8



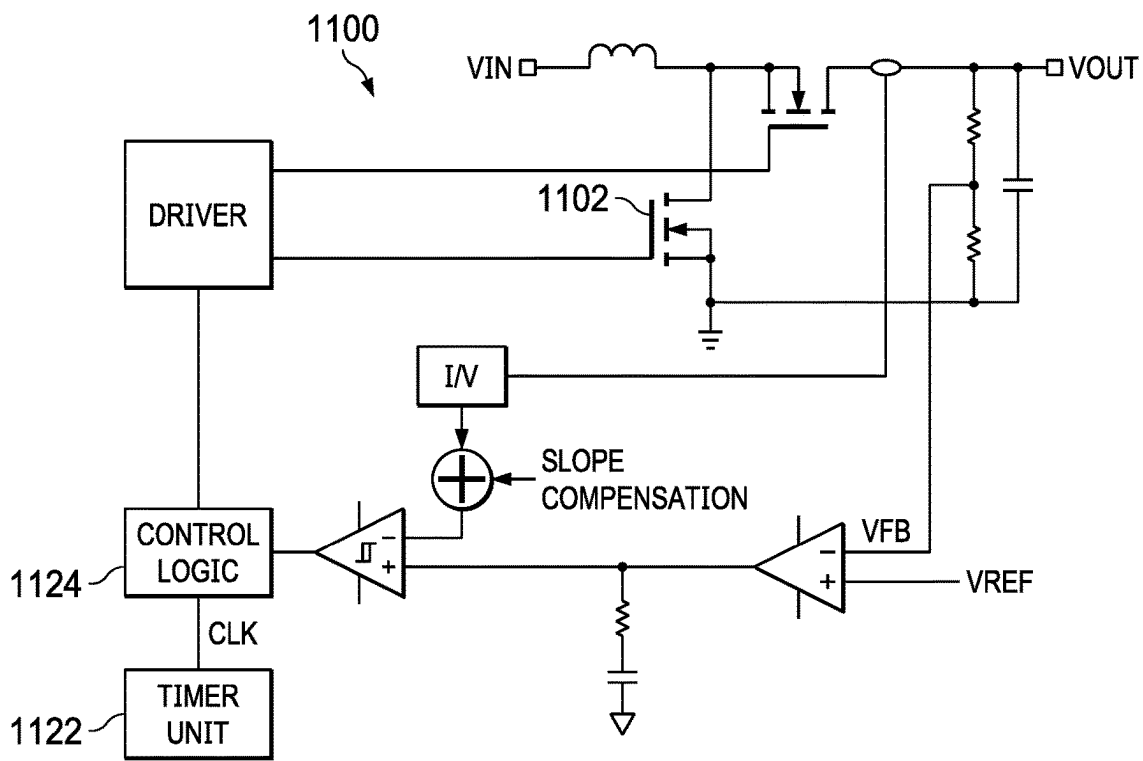


FIG. 11

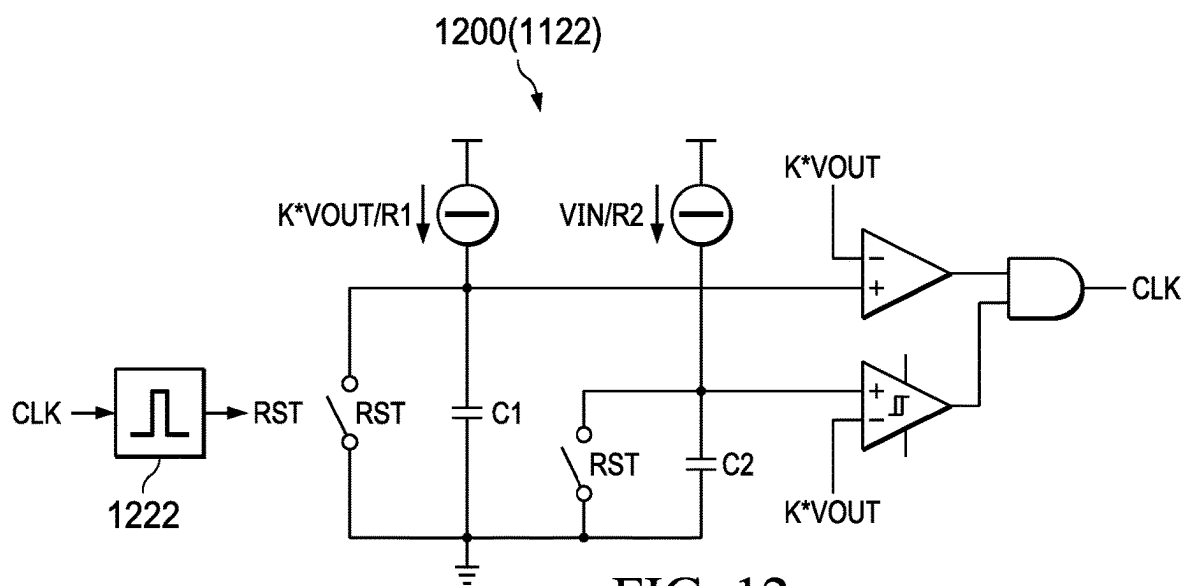


FIG. 12

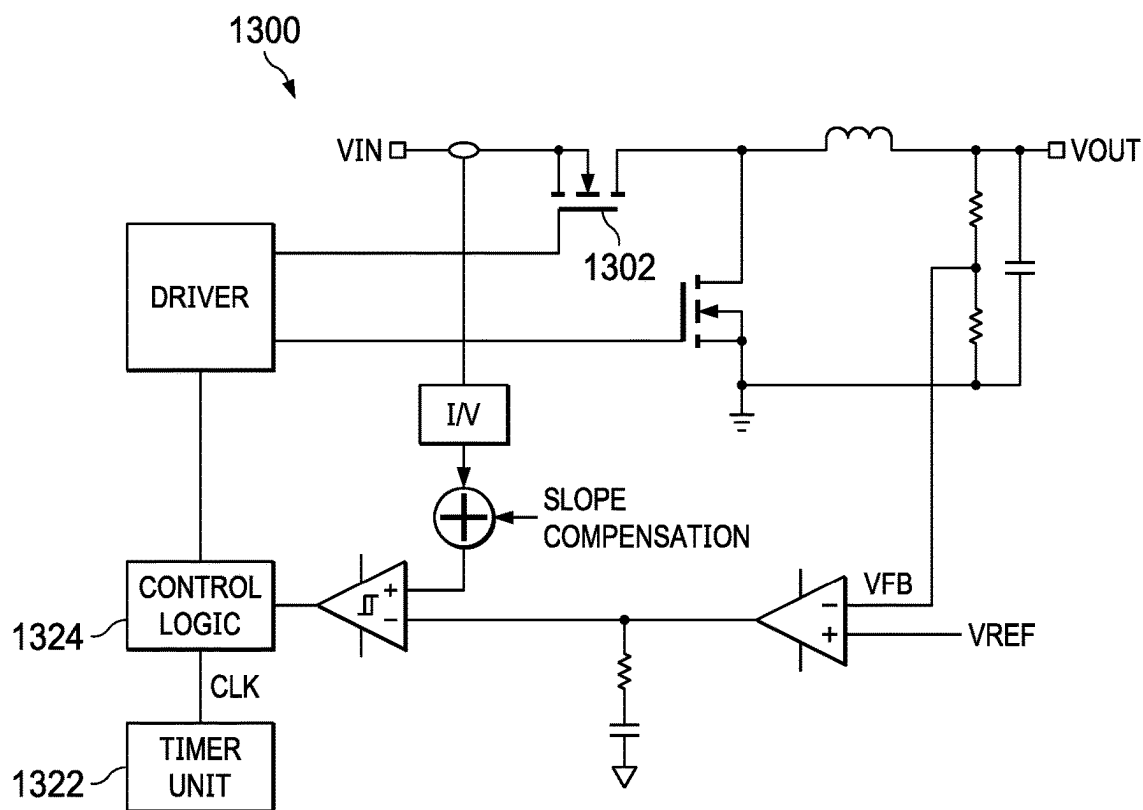


FIG. 13

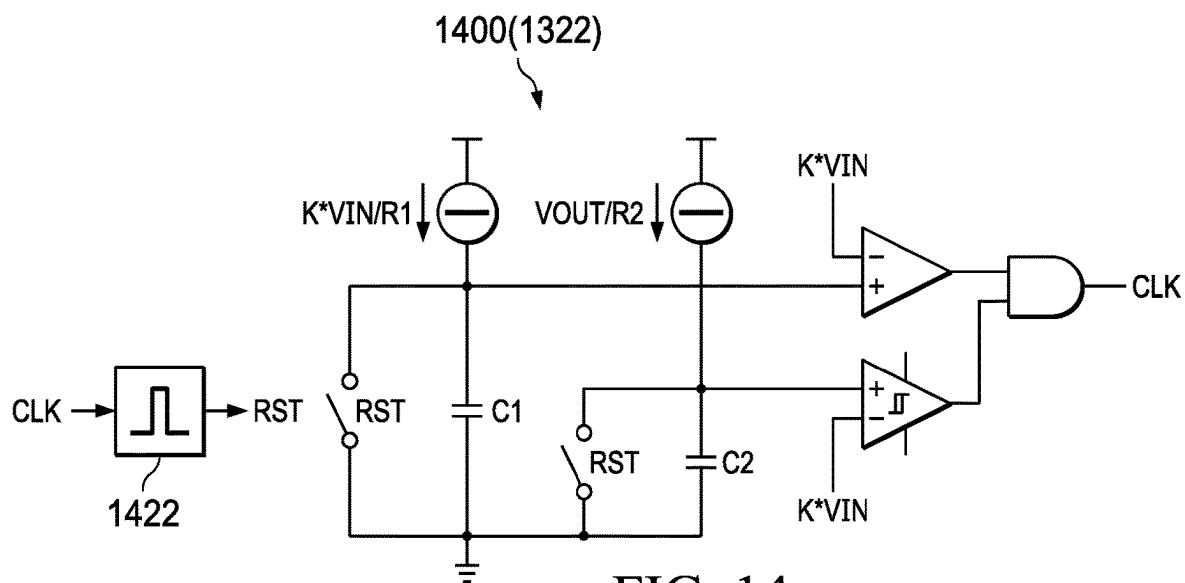


FIG. 14

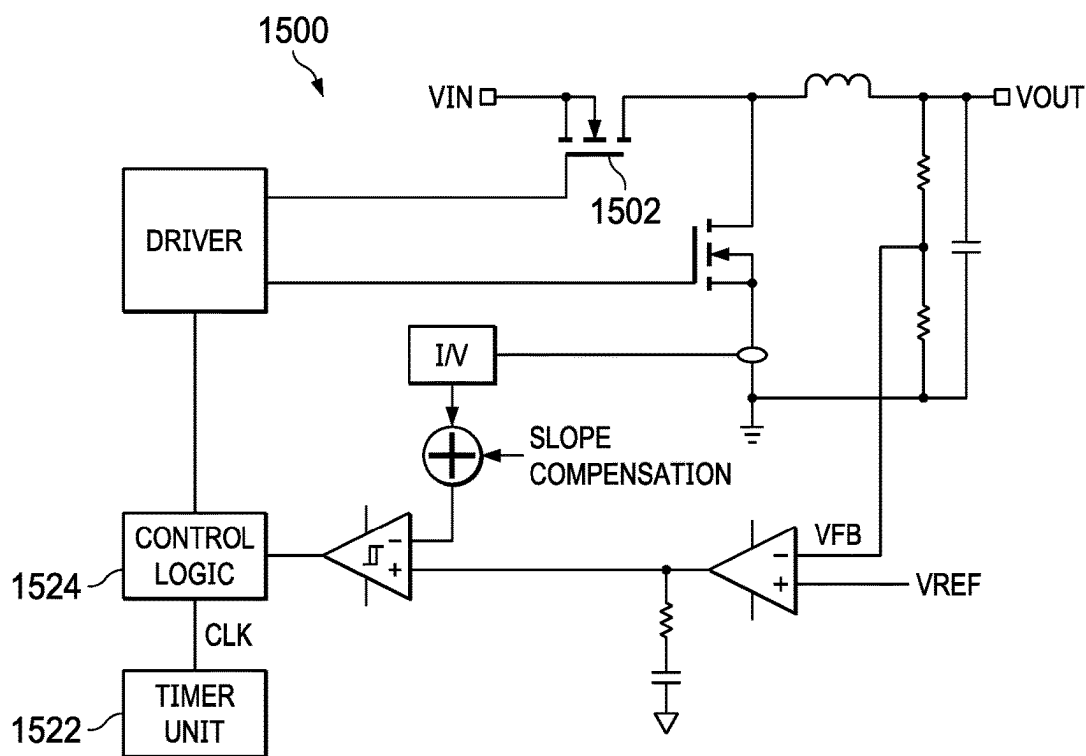


FIG. 15

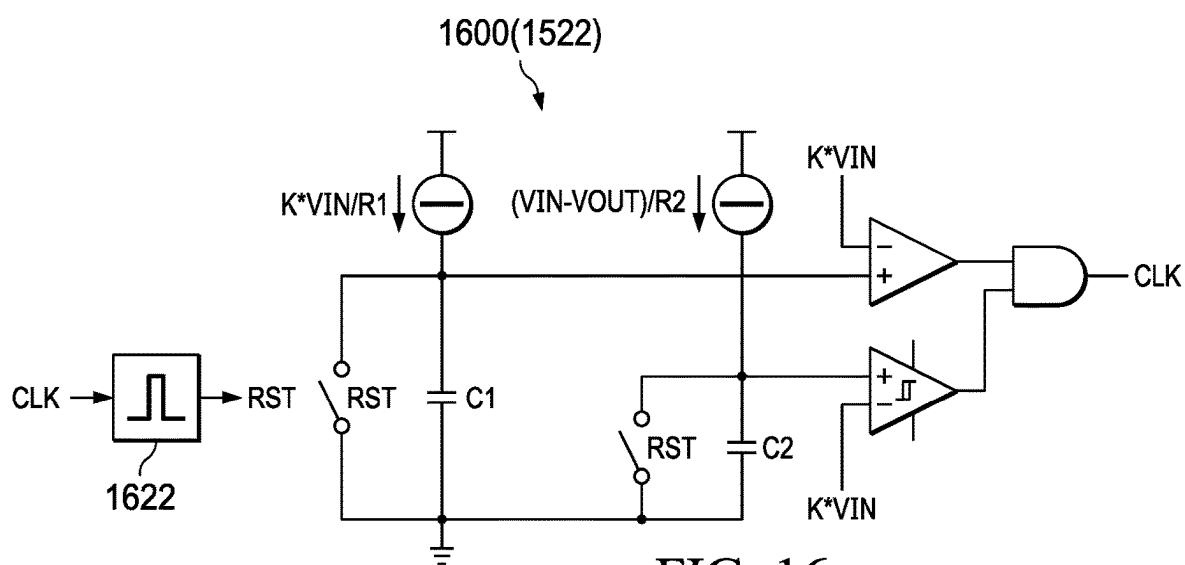


FIG. 16

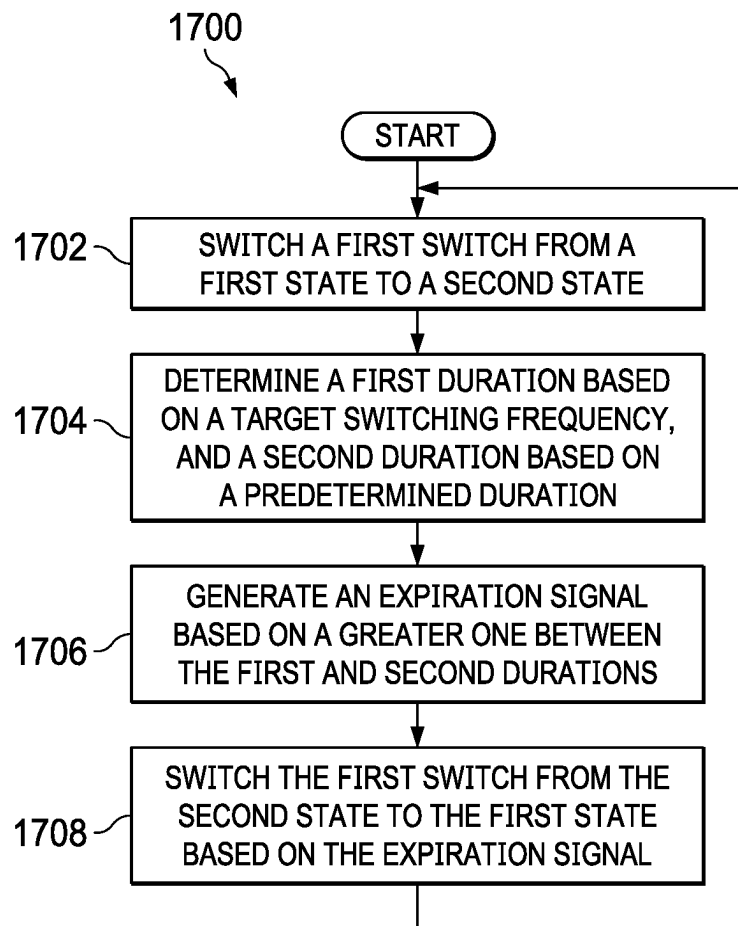


FIG. 17

1

ADAPTIVE OFF-TIME OR ON-TIME DC-DC CONVERTER

This application is a divisional of U.S. patent application Ser. No. 16/876,897 filed May 18, 2020, which is a continuation of PCT Application No. PCT/CN2020/070110 filed Jan. 2, 2020, both of which are incorporated herein by reference in their entirety.

TECHNICAL FIELD

This relates generally to integrated circuits, and more particularly to a current mode DC-DC converter system.

BACKGROUND

DC-DC converters are widely used to convert an input DC voltage to a desired output DC voltage to drive a load. A current mode DC-DC converter may include a current loop that determines on or off time of a switch in each switching cycle by sensing an inductor current flowing through an inductor that is coupled to a switch node of the DC-DC converter, thereby regulating the inductor current. In a conventional adaptive on-time or off-time current mode DC-DC converter, a pulse-width-modulation (PWM) signal that controls the switch is regulated based on the sensed inductor current, the on-time or off-time determined based on input and output voltages of the DC-DC converter. In a conventional fixed frequency current mode DC-DC converter, the PWM signal is regulated based on the sensed inductor current, and a clock signal with a fixed target frequency.

SUMMARY

This description relates generally to integrated circuits, and more particularly to a DC-DC converter system with a wider range of duty cycle. A DC-DC converter system, such as a switch mode DC-DC converter, usually includes a switch configured to operate between on and off states based on a frequency signal, such as a pulse-width-modulation (PWM) signal, to generate an output DC voltage to a load by periodically storing energy from a source that provides an input DC voltage in a magnetic field of an inductor or a transformer and releasing the energy from the magnetic field. The ratio between the output DC voltage and the input DC voltage is proportional to a duty cycle of the PWM signal.

In one example, this description provides a DC-DC converter system including a first switch coupled to a switching node and operable between first and second states, and a controller, coupled to the first switch, and configured to switch the first switch between the first and second states based on an input voltage and an output voltage of the DC-DC converter system. The controller includes: a timer unit including a first timer configured to determine a first duration based on a target switching frequency of the DC-DC converter system, a second timer configured to determine a second duration based on a predetermined duration substantially equal to or greater than a minimum duration of the first state of the first switch and the input and output voltages, and logic circuitry coupled to the first and second timers and configured to generate an expiration signal responsive to expiration of both the first and second durations; and a control logic unit configured to switch the first switch from the second state to the first state based on the expiration signal.

2

In another example, this description provides a controller for switching a first switch of a DC-DC converter system. The controller includes: a timer unit including a first timer configured to determine a first duration based on a target switching frequency of the DC-DC converter system, a second timer configured to determine a second duration based on input and output voltages of the DC-DC converter system and a predetermined duration substantially equal to or greater than a minimum duration of a first state of the first switch, and logic circuitry coupled to the first and second timers and configured to generate an expiration signal responsive to expiration of both the first and second durations; and a control logic unit configured to switch the first switch from a second state to the first state based on the expiration signal.

In yet another example, this description provides a DC-DC converter system including: a first switch coupled to a switching node of the DC-DC converter system and a voltage supply node, and operable between first and second states; a first timer including a first capacitive element with a first capacitance, a first timing switching coupled in parallel with the first capacitive element, a first current source coupled in series with the first capacitive element and configured to source or sink a first current to or from the first capacitive element, and a first comparator with a first input terminal coupled to the first capacitive element, a second input terminal configured to receive a first reference voltage, and an output terminal configured to generate a first timer expired signal responsive to expiration of a first duration determined based on the first capacitance, the first current source and the first reference voltage; a second timer including a second capacitive element with a second capacitance, a second timing switching coupled in parallel with the second capacitive element; a second current source coupled in series with the second capacitive element and configured to source or sink a second current to or from the second capacitive element, and a second comparator with a first input terminal coupled to the second capacitive element, a second input terminal configured to receive a second reference voltage, and an output terminal configured to generate a second timer expired signal responsive to expiration of a second duration determined based on the second capacitance, the second current source and the second reference voltage; a logic gate coupled to outputs of the first and second comparators, and configured to generate an expiration signal based on expiration of both the first and second durations; and a control logic unit, coupled between the logic gate and a control terminal of the first switch, configured to switch the first switch from the second state to the first state based on the expiration signal.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a schematic block diagram of a DC-DC converter system in accordance with a first implementation of this description.

FIGS. 2A-2D show schematic circuit diagrams of a timer unit of the DC-DC converter system of FIG. 1.

FIG. 3 is a schematic block diagram of a DC-DC converter system in accordance with a second implementation of this description.

FIG. 4 is a schematic circuit diagrams of a timer unit of the DC-DC converter system of FIG. 3.

FIG. 5 is a schematic block diagram of a DC-DC converter system in accordance with a third implementation of this description.

FIG. 6 is a schematic circuit diagrams of a timer unit of the DC-DC converter system of FIG. 5.

FIG. 7 is a schematic block diagram of a DC-DC converter system in accordance with a fourth implementation of this description.

FIG. 8 is a schematic circuit diagrams of a timer unit of the DC-DC converter system of FIG. 7.

FIG. 9 is a schematic block diagram of a DC-DC converter system in accordance with a fifth implementation of this description.

FIG. 10 is a schematic circuit diagrams of a timer unit of the DC-DC converter system of FIG. 9.

FIG. 11 is a schematic block diagram of a DC-DC converter system in accordance with a sixth implementation of this description.

FIG. 12 is a schematic circuit diagrams of a timer unit of the DC-DC converter system of FIG. 11.

FIG. 13 is a schematic block diagram of a DC-DC converter system in accordance with a seventh implementation of this description.

FIG. 14 is a schematic circuit diagrams of a timer unit of the DC-DC converter system of FIG. 13.

FIG. 15 is a schematic block diagram of a DC-DC converter system in accordance with an eighth implementation of this description.

FIG. 16 is a schematic circuit diagrams of a timer unit of the DC-DC converter system of FIG. 15.

FIG. 17 is a flow chart of a method of operating a DC-DC converter system in accordance with an implementation of this description.

DETAILED DESCRIPTION

This description relates to current mode DC-DC converter systems.

Referring to FIG. 1, a schematic block diagram of a DC-DC converter system 100 in accordance with a first implementation of this description is shown. More particularly, FIG. 1 shows an adaptive off-time current mode boost DC-DC converter system with peak current control topology.

The system 100 includes a first switch 102 coupled between a switch node SW and a voltage supply node, for example, a ground node GND, and a second switch 104 coupled between the switch node SW and an output node VOUT of the system 100, thereby allowing a current flowing from the switch node SW to the output node VOUT. The first and second switches 102 and 104, also named respectively as low side and high side switches, can be transistors, for example, N-channel MOSFETs that are respectively controlled by gate drive signals LSD_ON and HSD_ON to alternately operable between first and second states, e.g. on and off states, allowing a current to follow from the switch node SW towards the voltage supply node GND, and from the switch node SW towards the output node VOUT. In an alternate example, the second switch 104 can be replaced by a diode that allows current to flow from the switch node SW to the output node VOUT in a unidirectional manner. The system 100 also includes an input inductor 106 coupled between an input node VIN and the switch node SW, and an output capacitive element 108 coupled between the output node VOUT and the ground node GND.

The system 100 includes a controller 110 coupled to the first and second switches 102 and 104 to generate a PWM signal to alternately switch on and off the first and second switches 102 and 104 through a driver unit 112 which generates the gate drive signals LSD_ON and HSD_ON

based on the PWM signal. In a preferred example, the driver unit 112 can be either a part of or separate from the controller 110.

The controller 110 includes a sensing unit 114 configured to generate a control signal Sc based on a difference between a sensing voltage Vs proportional to an inductor current IL through the inductor 106 and a control voltage Vc proportional to a difference between a reference voltage VREF and a feedback voltage VFB proportional to the output voltage VOUT, generated by an amplifier 116. In one example, the sensing unit 114 includes a comparator 118 configured to generate the control signal Sc to switch the first switch 102 from the on state to the off state if the sensing voltage Vs increases to the control voltage Vc. In one example, the sensing voltage Vs is proportional to a current Is flowing through the first switch 102 and obtained through a current-to-voltage (I/V) unit 120, for example, by sensing a voltage across a sensing resistor (not shown) coupled between the first switch 102 and the ground node GND.

The controller 110 includes a control logic 124 configured to switch the first switch 102 from the on state to the off state through the driver unit 112, based on the control signal Sc. However, a minimum duration of the on state, also known as a minimum on-time Ton_min, of the first switch is usually limited due to various factors of the DC-DC converter system 100, such as blanking time of inductor current IL sensing, delay caused by the comparator 118, the control logic 124 and/or the driver unit 112. The minimum on-time of the first switch 102 limits the range of a ratio of output voltage VOUT to the input voltage VIN.

The controller 110 also includes a timer unit 122 configured to determine a preferred duration of the off state, also known as an off-time Toff, of the first switch 102 based on a target switching frequency fsw of the DC-DC converter system 100, the minimum on-time Ton_min of the first switch, and the input and output voltages VIN and VOUT of the system 100.

The timer unit 122 is further configured to generate an expiration signal ST to switch the first switch 102 from the off state to the on state when the preferred off-time Toff expires. The control logic 124 is configured to generate the PWM signal based on the control signal Sc and the expiration signal ST. For example, the control logic 124 can be an edge-triggered SR flip flop that asserts the PWM signal based on the expiration signal ST and de-asserts the PWM signal based on the control signal Sc.

FIG. 2A shows an example schematic circuit diagram of a timer unit 200, such as the timer unit 122 of the DC-DC converter system 100 of FIG. 1. The timer unit 200 includes a first timer 202 configured to generate a first timer expired signal ST1 based on a first duration T1. In one example, the first duration T1 is a nominal duration of the second state, e.g. the off state, of the first switch 102 determined based on the target switching frequency fsw of the DC-DC converter system 100 and the input and output voltages VIN and VOUT, such that the DC-DC converter system operating at the target switching frequency fsw converts the input voltage VIN to the output voltage VOUT by keeping the first switch 102 at the off state for the nominal duration in each target switching cycle T. In the example of the adaptive off-time current mode boost DC-DC converter system with peak current control topology, the first duration T1 is determined in accordance with the equation below:

$$T1 = T \cdot \frac{VIN}{VOUT} \quad (1)$$

where T is the target switching cycle, T=1/fsw.

5

In one example, the first timer **202** includes a first capacitive element **204** with a capacitance $C1$. The first capacitive element **204** is coupled in parallel with a first charging control switch **206** that is controlled by the gate drive signal LSD_ON , and in series with a first current source **208** configured to source a first charging current $Ic1$ to the first capacitive element **204**. In the example of FIG. 2, $Ic1 = VOUT/R1$. Charging the first capacitive element **204** is triggered responsive to the first switch **102** being switched from the on state to the off state. The first timer **202** includes a first comparator **210** with an inverting input coupled to a reference voltage generator shown in FIG. 2B to receive a reference voltage $Vref1 = K \cdot VIN$. The first capacitive element **204** is coupled between a non-inverting input of the first comparator **210** and the ground node GND. The first capacitive element **210** is configured to generate a first timer expired signal S_{T1} when a voltage across the first capacitive element **204** increased to the reference voltage $K \cdot VIN$. Accordingly, the first duration $T1$ is determined by the first timer **202** in accordance with the equation below:

$$T1 = K \cdot R1 \cdot C1 \cdot \frac{VIN}{VOUT} \quad (2)$$

where K is a number greater than 0, $K \cdot R1$ is resistance of a resistor **236** of a charging path of the first timer **202** shown in FIG. 2C, and $K \cdot R1 \cdot C1$ is configured to be substantially equal to the target switching cycle $T = 1/fsw$ of the DC-DC converter system **100** within acceptable error range resulting from inherent errors of the first capacitive element **210** and the resistor **236**. However, the first charging current $Ic1$ and the reference voltage $Vref1$ can be other values as long as meeting the equation below:

$$T1 = \frac{Vref2}{Ic1} \cdot C1 = T \cdot \frac{VIN}{VOUT} \quad (3)$$

The timer unit **200** also includes a second timer **212** with a structure similar to that of the first timer **202** except that a second capacitive element **214** of the second timer **212** has a capacitance $C2$ and is charged by a second current source **216** with a second charging current $Ic2$, where $Ic2 = (VOUT - VIN)/R2$. The second timer **212** is configured to be triggered substantially simultaneously with the first timer based on the gate drive signal LSD_ON , and to generate a second timer expired signal S_{T2} based on a second duration $T2$ that is determined based on the minimum on-time Ton_min of the first switch **102** and the input and output voltages VIN and $VOUT$. The second duration $T2$ is provided in accordance with the equations below:

$$T2 = K \cdot R2 \cdot C2 \cdot \frac{VIN}{VOUT - VIN} \quad (4)$$

where $K \cdot R2$ is resistance of a resistor **242** of a charging path of the second timer **212** shown in FIG. 2D, and $K \cdot R2 \cdot C2$ is configured to be substantially equal to or slightly greater, e.g. 10 ns greater, than the minimum on-time Ton_min of the first switch **102**.

In one example, the second timer **212** includes a second charging switch **220** configured to be switched off simultaneously with switching off the first charging switch **206**, and

6

$K \cdot R2 \cdot C2$ is configured to be substantially equal to the minimum on-time Ton_min of the first switch **102**, such that the second duration equals an off-time of the first switch **102** determined under the condition that the on-time of the first switch **102** is the minimum on-time Ton_min .

In another example, the second timer **212** includes a second charging switch **220** configured to be switched off simultaneously with switching off the first charging switch **206**, and $K \cdot R2 \cdot C2$ is configured to be slightly, e.g. 10 ns, greater, than the minimum on-time Ton_min of the first switch **102** to ensure the second duration longer than an off-time of the first switch **102** determined under the condition that the on-time of the first switch **102** is the minimum on-time Ton_min .

In yet another example, $K \cdot R2 \cdot C2$ is configured to be substantially equal to the minimum on-time Ton_min of the first switch **102**, and the second timer **212** further includes a delay unit **222** such that the second charging switch **220** is configured to be switched off slightly later, e.g. 50 ns or more, than switching off the first charging switch **206** to ensure the second duration longer than an off-time of the first switch **102** determined under the condition that the on-time of the first switch **102** is the minimum on-time Ton_min .

In one example, the second timer **212** further includes a second comparator **224** with an inverting input coupled to another reference voltage generator to receive another reference voltage $Vref2$. The second charging current $Ic2$ and the reference voltage $Vref2$ can be other values as long as meeting the equation below:

$$T2 = \frac{Vref2}{Ic2} \cdot C1 \geq Ton_min \cdot \frac{VIN}{VOUT - VIN} \quad (5)$$

The timer unit **200** also includes a logic gate **218** configured to generate the expiration signal S_T responsive to both the first and second timer expired signals S_{T1} and S_{T2} being asserted. In one example, when $VOUT/(R1 \cdot C1) > (VOUT - VIN)/(R2 \cdot C2)$, the second duration $T2$ is smaller than the first duration $T1$, the preferred duration of the off state of the first switch **102** is determined by the first duration $T1$, which is the nominal duration of the off state of the first switch **102** determined based on the target switching frequency fsw and the input and output voltages of the system **100**. In another example, when $VOUT/(R1 \cdot C1) < (VOUT - VIN)/(R2 \cdot C2)$, the preferred duration of the off state of the first switch **102** is determined by the second duration $T2$, and the duration of the on state of the first switch **102** is regulated at $(VOUT - VIN)/VIN \cdot Toff = K \cdot R2 \cdot C2$, which is substantially equal to or greater than the minimum on-time Ton_min of the first switch **102**, therefore the inductor current IL can still be regulated based on the sensing voltage Vs . In such situation, an actual switching cycle of the DC-DC converter system **100** is configured to be $K \cdot R2 \cdot C2 \cdot VOUT/(VOUT - VIN)$, with is greater than the target switching cycle of the DC-DC converter system **100**.

FIG. 2B shows an example schematic circuit diagram of the reference voltage generator **226** that provides the reference voltage $K \cdot VIN$. In one example, the reference voltage generator **226** includes a voltage divider **228** generating the reference voltage $K \cdot VIN$ proportional to the input voltage VIN .

FIG. 2C shows an example schematic circuit diagram of a current source **230**, for example, the first current source **208** of the timer unit **200** of FIG. 2. The first current source **230** includes an error amplifier **232** having an output termi-

nal coupled to a gate node of a transistor **234**, a non-inverting input terminal configured to receive a reference voltage $K \cdot V_{OUT}$ which can be provided in a similar manner as the reference voltage generator **220** shown in FIG. 2B, and an inverting input terminal coupled to a source node of the transistor **234**. The first current source **230** also includes the resistor **236** coupled between the source node of the transistor **234** and the ground node GND and having a resistance of $K \cdot R1$, and a current mirror **238** coupled to a drain node of the transistor **234** and configured to mirror a current flowing through the resistor **236**, which is provided as the first charging current $I_{c1} = V_{OUT}/R1$.

FIG. 2D shows an example schematic circuit diagram of another current source **240**, for example, the second current source **216** of the timer unit **200** of FIG. 2. The second current source **240** has a structure similar to that of the first current source **230** except that a voltage difference across the resistor **242** is configured to be $V_{OUT} - V_{IN}$. The current mirror **244** is configured to mirror a current flowing through the resistor **242**, which is provided as the second charging current $I_{c2} = (V_{OUT} - V_{IN})/R2$.

Referring back to FIG. 1, the control logic unit **124** is configured to switch the first switch **102** from the off state to the on state based on the expiration signal S_T . Therefore, the duration of the off state, i.e., the off time T_{off} , of the first switch **102** is configured to be a greater one between the first and second duration $T1$ and $T2$.

In a conventional adaptive off-time current mode boost DC-DC converter system, the off-time T_{off} of the first switch, e.g. the low side switch, is configured to be a nominal off-time T_{off}^* determined in accordance with the equation below:

$$T_{off} = T \cdot V_{IN}/V_{OUT} \quad (6)$$

Due to the limit of the minimum on-time T_{on_min} of the system, an on duty cycle range of the system is limited between T_{on_min}/T and 1, and thus a ratio of V_{OUT} to V_{IN} range is limited between $T/(T - T_{on_min})$ and ∞ . Similarly, operation ranges of other conventional adaptive on-time/off-time current mode DC-DC converter systems with other topologies are also limited by the nominal off-time and minimum on-time of the system, or a nominal on-time and a minimum-off time of the system. Table 1 lists the operation ranges of conventional adaptive on-time/off-time current mode DC-DC converter systems with different topologies.

TABLE 1

Operation ranges of conventional adaptive on-time/off-time current mode DC-DC converter systems			
Buck/ Boost	Topology	On Duty Cycle Range	V_{OUT}/V_{IN} range
Boost	Peak Current + adaptive Toff	Current Regulated, $T_{on} \geq T_{on_min}$ $T_{off} = T * V_{IN}/V_{OUT}$	(T_{on_min}/T , 1) ($T/(T - T_{on_min})$, ∞)
	Valley current + adaptive Ton	$T_{on} = T * (V_{OUT} - V_{IN})/V_{OUT}$ (0, 1 - T_{off_min}/T) Current Regulated, $T_{off} \geq T_{off_min}$	(1, T/T_{off_min})
Buck	Peak Current + adaptive Toff	Current Regulated, $T_{on} \geq T_{on_min}$ $T_{off} = T * (V_{IN} - V_{OUT})/V_{IN}$	(T_{on_min}/T , 1) (T_{on_min}/T , 1)
	Valley current + adaptive Ton	$T_{on} = T * V_{OUT}/V_{IN}$ (0, 1 - T_{off_min}/T) Current Regulated, $T_{off} \geq T_{off_min}$	(0, 1 - T_{off_min}/T)

In this description, the proposed adaptive off-time current mode boost DC-DC converter system **100** dynamically extends the off-time T_{off} of the first switch **102** when the off-time determined based on the minimum on-time T_{on_min} and the input and output voltages V_{IN} and V_{OUT} is greater than the nominal off-time of the DC-DC converter system. As the off-time of the first switch **102** can be extended to $T_{on_min} \cdot V_{IN}/(V_{OUT} - V_{IN})$ when a target ratio of V_{OUT} to V_{IN} is less than $T/(T - T_{on_min})$, the range of on duty cycle can be extended between 0 and 1, and the range of V_{OUT}/V_{IN} can be extended between 1 and ∞ .

FIG. 3 shows a schematic block diagram of a DC-DC converter system **300** in accordance with a second implementation of this description. More particularly, FIG. 3 shows an adaptive on-time current mode boost DC-DC converter system with valley current control topology.

The DC-DC converter system **300** is substantially similar to the DC-DC converter system **100** of FIG. 1 except that the sensing voltage V_s is generated proportional to a current flowing through the second switch **304**, i.e. the high side switch, the comparator **318** is configured to generate the control signal S_c when the sensing voltage V_s decreases to the control voltage V_c , and the control logic **324** is configured to switch on the first switch **302** based on the control signal S_c and to switch off the first switch **302** based on the expiration signal S_T generated by the timer unit **322**.

FIG. 4 shows an example schematic circuit diagram of a timer unit **400**, such as the timer unit **322** of the DC-DC converter system **300** of FIG. 3. The timer unit **400** includes a first timer **402** configured to generate a first timer expired signal S_{T1} based on a first duration $T1$. In one example, the first duration $T1$ is a nominal duration of the on state, i.e. a nominal on-time T_{on} , of the first switch **302** determined based on a target switching frequency f_{sw} of the DC-DC converter system **300** and input and output voltages V_{IN} and V_{OUT} in accordance with the equation below:

$$T1 = T \cdot \frac{V_{OUT} - V_{IN}}{V_{OUT}} \quad (6)$$

where T is the target switching cycle of the system **300**, $T = 1/f_{sw}$.

The first timer **402** is configured to determine the first duration **T1**, i.e., the nominal on-time of the first switch **302**, in accordance with the equation below:

$$T1 = K \cdot R1 \cdot C1 \cdot \frac{VOUT - VIN}{VOUT} \quad (7)$$

where $K \cdot R1 \cdot C1$ is configured to be substantially equal to a target switching cycle $T=1/fsw$ of the DC-DC converter system **300**.

The timer unit **400** also includes a second timer **412** with a structure similar to that of the first timer **402** except that the second timer **412** is configured to generate a second timer expired signal **ST2** based on a second duration **T2** provided in accordance with the equations below:

$$T2 = K \cdot R2 \cdot C2 \cdot \frac{VOUT - VIN}{VOUT} \quad (8)$$

where $K \cdot R2 \cdot C2$ is configured to be substantially equal to or slightly greater, e.g. 10 ns greater, than a minimum duration of the off state, i.e., the minimum off-time $Toff_min$, of the first switch **302**.

The first and second timer **402** and **412** are configured to start timing responsive to the second switch **304** being switched from the on state to the off state, i.e., when the first switch **302** is switched from the off state to the on state. Similar to the timer unit **200** of FIG. 2, the timer unit **400** is configured to generate an expiration signal S_T responsive to both of the first and second timer expired signals S_{T1} and S_{T2} being asserted.

Similar to the DC-DC converter system **100** of FIG. 1, as the on-time of the first switch **302** can be extended to $Toff_min \cdot (VOUT - VIN)/VIN$ when a target ratio of $VOUT$ to VIN is greater than $T/Toff_min$, the range of on duty cycle can be extended between 0 and 1, and the range of $VOUT/VIN$ can be extended between 1 and ∞ .

FIG. 5 in combination with FIG. 6 shows a schematic block diagram of a DC-DC converter system **500** in accordance with a third implementation of this description. More particularly, FIG. 5 shows an adaptive on-time current mode buck DC-DC converter system with valley current control topology. Similar to the adaptive on-time current mode boost DC-DC converter system **300** shown in FIG. 3 in combination with the timer unit **400** of FIG. 4, the DC-DC converter system **500** is configured to switch off the first switch **502**, i.e. the high side switch, based on an expiration signal S_T generated by the timer unit **522**. The timer unit **522**, shown as the timer unit **600** of FIG. 6 in more detail, is configured to generate the expiration signal S_T to switch

off the first switch **502** responsive to expiration of both of the first and second duration **T1** and **T2** respectively determined by the first and second timers. The first duration **T1** is determined based on a nominal duration of the on state, e.g. a nominal on-time Ton , of the first switch **602** determined based on the switching frequency fsw and the input and output voltages VIN and $VOUT$ of the DC-DC converter system **500**, and the second duration **T2** is determined based on the input and output voltages VIN and $VOUT$ and a predetermined duration substantially equal to greater than the minimum duration of the off state, also known as minimum off-time $Toff_min$, of the first switch **502**. Similar to the DC-DC converter system **300** of FIG. 3, as the duration of the on state, i.e., the on-time, of the first switch **502** can be extended to $Toff_min \cdot VOUT/(VIN - VOUT)$ when a target ratio of $VOUT$ to VIN is greater than $1 - Toff_min/T$, the range of on duty cycle can be extended between 0 and 1, and the range of $VOUT/VIN$ can be extended between 0 and 1.

FIG. 7 in combination with FIG. 8 shows a schematic block diagram of a DC-DC converter system **700** in accordance with a fourth implementation of this description. More particularly, FIG. 7 shows an adaptive off-time current mode buck DC-DC converter system with peak current control topology. Similar to the adaptive on-time current mode buck DC-DC converter system **500** shown in FIG. 5 in combination with the timer unit **600** of FIG. 6, the DC-DC converter system **700** is configured to switch on the first switch **702**, i.e. the high side switch, based on the expiration signal S_T generated by the timer unit **722**. The timer unit **722**, shown as the timer unit **800** of FIG. 8 in more detail, is configured to generate the expiration signal S_T to switch on the first switch **702** when both of the first and second duration **T1** and **T2** respectively determined by the first and second timers expire. The first duration **T1** is determined based on a nominal duration of the off state, i.e. a nominal off-time, of the first switch **702** which is determined based on the switching frequency fsw and the input and output voltages VIN and $VOUT$ of the DC-DC converter system **700**, and the second duration **T2** is determined based on the input and output voltages VIN and $VOUT$ and a predetermined duration substantially equal to greater than the minimum on-time Ton_min of the first switch **702**. Similar as the DC-DC converter system **500** of FIG. 5, as the off-time of the first switch **702** can be extended to $Ton_min \cdot (VIN - VOUT)/VOUT$ when a target ratio of $VOUT$ to VIN is less than Ton_min/T , the range of on duty cycle can be extended between 0 and 1, and the range of $VOUT/VIN$ can be extended between 0 and 1.

Table 2 lists the operation ranges of adaptive on-time/off-time current mode DC-DC converter systems with different topologies in accordance with the first to fourth implementations of this description.

TABLE 2

Operation ranges of adaptive on-time/off-time current mode DC-DC converter systems in accordance with implementations of this description				
Buck/Boost	Topology	Ton and Toff	On Duty Cycle Range	VOUT/VIN range
Boost	Peak Current + adaptive Toff	Current Regulated, $Ton \geq Ton_min$ $Toff = \max\{T * VIN/VOUT, Ton_min * VIN/(VOUT - VIN)\}$	(0, 1)	(1, ∞)
	Valley current + adaptive Ton	$Ton = \max\{T * (VOUT - VIN)/VOUT, Toff_min * (VOUT - VIN)/VIN\}$ Current Regulated, $Toff \geq Toff_min$	(0, 1)	(1, ∞)

TABLE 2-continued

Operation ranges of adaptive on-time/off-time current mode DC-DC converter systems in accordance with implementations of this description				
Buck/ Boost	Topology	Ton and Toff	On Duty Cycle Range	VOUT/VIN range
Buck	Peak Current + adaptive Toff	Current Regulated, $T_{on} \geq T_{on_min}$ $T_{off} = \max\{T * (VIN - VOUT)/VIN,$ $T_{on_min} * (VIN - VOUT)/VOUT\}$	(0, 1)	(0, 1)
	Valley current + adaptive Ton	$T_{on} = \max\{T * VOUT/VIN,$ $T_{off_min} * VOUT/(VIN - VOUT)\}$ Current Regulated, $T_{off} \geq T_{off_min}$	(0, 1)	(0, 1)

15

Referring to FIG. 9, a schematic block diagram of a DC-DC converter system 900 in accordance with a fifth implementation of this description is shown. More particularly, FIG. 9 shows a fixed frequency current mode boost DC-DC converter system with peak current control topology. The DC-DC converter system 900 is substantially similar to the DC-DC converter system 100 of FIG. 1 except that the sensing voltage V_s provided to the comparator 918 is proportional to a combination of the current flowing through the first switch 902 and a slope compensation signal, and the control logic 924 is configured to switch on the first switch 902 based on a clock signal CLK generated by the timer unit 922, for example, a rising edge of the clock signal CLK.

FIG. 10 shows an example schematic circuit diagram of a timer unit 1000, such as the timer unit 922 of the DC-DC converter system 900 of FIG. 9. The timer unit 1000 is substantially similar to the timer unit 200 of FIG. 2, except that the timer unit 922 further includes a one-shot signal generator 1022 coupled to the switches 1006 and 1020 to provide a one-shot signal RST based on the clock signal CLK, such that the first and second timers 1002 and 1012 start timing responsive to the first switch 902 being switched from the off state to the on state. The first timer 1002 is configured to generate a first timer expired signal S_{T1} based on a first duration T1. In one example, the first duration T1 is a target switching cycle $T=1/fsw$, where fsw is a target switching frequency of the DC-DC converter system 900. The first duration T1 is provided in accordance with the equations below:

$$T1=K \cdot R1 \cdot C1 \quad (9)$$

where $K \cdot R1 \cdot C1$ is configured to be substantially equal to the target switching cycle $T=1/fsw$ of the DC-DC converter system 900.

The timer unit 1000 also includes a second timer 1012 configured to generate a second timer expired signal S_{T2} based on a second duration T2 that is determined based on a minimum duration of the on state, i.e., a minimum on-time

T_{on_min} , of the first switch 902 and the input and output voltages VIN and VOUT. In one example, the second duration T2 is determined based on the input and output voltages VIN and VOUT, and a predetermined duration substantially equal to greater than the minimum on-time T_{on_min} of the first switch 902. The second duration T2 is provided in accordance with the equations below:

$$T2 = K \cdot R2 \cdot C2 \cdot \frac{VOUT - VIN}{VOUT} \quad (10)$$

where $K \cdot R2 \cdot C2$ is configured to be substantially equal to or slightly greater, e.g. 10 ns greater, than the minimum on-time T_{on_min} of the first switch 902.

The timer unit 1000 further includes a gate logic 1018 configured to generate, for example, a rising edge, of the clock signal CLK based on the first and second timer expired signals S_{T1} and S_{T2} , wherein the cycle of the clock signal CLK is configured to be the larger one of the first and second duration T1 and T2.

In a conventional fixed frequency peak current control mode boost DC-DC converter system, a cycle of a clock signal CLK that periodically switches the first switch from a second state, e.g. the off state, to a first state, e.g. the on state, is fixed by the target switching frequency fsw of the conventional system.

Due to the minimum on-time T_{on_min} of the conventional system, the on duty cycle range of the system is limited between T_{on_min}/T and $1 - T_{off_min}/T$, and thus a range of a ratio of VOUT to VIN is limited between $T/(T - T_{on_min})$ and T/T_{off_min} . Similarly, operation ranges of other conventional fixed frequency current mode DC-DC converter systems with other topologies are also limited by the target switching cycle and minimum on or off time of the first switch of the system. Table 3 lists the operation ranges of conventional fixed frequency current mode DC-DC converter systems with different topologies.

TABLE 3

Operation ranges of conventional fixed frequency current mode DC-DC converter systems				
Buck/ Boost	Topology	Ton and Toff	On Duty Cycle Range	VOUT/VIN range
Boost	Fixed frequency + Peak Current	Current Regulated, $T_{on} \geq T_{on_min}$ Fixed period T	$(T_{on_min}/T,$ $1 - T_{off_min}/T)$	$(T/(T - T_{on_min}),$ $T/T_{off_min})$
	Fixed frequency + Valley Current	Current Regulated, $T_{off} \geq T_{off_min}$ Fixed Period T	$(T_{on_min}/T,$ $1 - T_{off_min}/T)$	$(T/(T - T_{on_min}),$ $T/T_{off_min})$

TABLE 3-continued

Operation ranges of conventional fixed frequency current mode DC-DC converter systems				
Buck/ Boost	Topology	Ton and Toff	On Duty Cycle Range	VOUT/VIN range
Buck	Fixed frequency + Peak Current	Current Regulated, $T_{on} \geq T_{on_min}$ Fixed period T	$(T_{on_min}/T,$ $1 - T_{off_min}/T)$	$(T_{on_min}/T,$ $1 - T_{off_min}/T)$
	Fixed frequency + Valley Current	Current Regulated, $T_{off} \geq T_{off_min}$ Fixed Period T	$(T_{on_min}/T,$ $1 - T_{off_min}/T)$	$(T_{on_min}/T,$ $1 - T_{off_min}/T)$

In this description, the proposed fixed frequency current mode boost DC-DC converter system **900** dynamically extends the switching cycle when a switching cycle determined based on the minimum on-time T_{on_min} and the input and output voltages VIN and VOUT is greater than the target switching cycle of the DC-DC converter system. As the switching cycle can be extended to $T_{on_min} \cdot VOUT / (VOUT - VIN)$ when a target ratio of VOUT to VIN is less than $T / (T - T_{on_min})$, the range of on duty cycle can be extended between 0 and $1 - T_{off_min}/T$, and the range of VOUT/VIN can be extended between 1 and T / T_{off_min} .

FIG. **11** in combination with FIG. **12** shows a schematic block diagram of a DC-DC converter system **1100** in accordance with a sixth implementation of this description. More particularly, FIG. **11** shows a fixed current mode boost DC-DC converter system with valley current control topology. Similar to the timer unit **1000** of FIG. **10**, the timer unit **1200** is configured to generate a clock signal CLK with a cycle determined based on a larger one between the target switching cycle of the DC-DC converter system **1100** and an adjusted cycle determined based on the input and output voltages VIN and VOUT and a predetermined duration substantially equal to greater than the minimum off time T_{off_min} of the first switch **1102** of the DC-DC converter system **1100**. The control logic **1124** is configured to switch off the first switch **1102** responsive to, for example, a rising edge, of the clock signal CLK.

FIG. **13** in combination with FIG. **14** shows a schematic block diagram of a DC-DC converter system **1300** in accordance with a seventh implementation of this description. More particularly, FIG. **13** shows a fixed current mode

buck DC-DC converter system with peak current control topology. Similar to the timer unit **1000** of FIG. **10**, the timer unit **1400** is configured to generate a clock signal CLK with a cycle determined based on a larger one between the target switching cycle of the DC-DC converter system **1300** and an adjusted cycle determined based on the input and output voltages VIN and VOUT and a predetermined duration substantially equal to greater than the minimum on time T_{on_min} of the first switch **1302** of the DC-DC converter system **1300**. The control logic **1324** is configured to switch on the first switch **1302** responsive to, for example, a rising edge, of the clock signal CLK.

FIG. **15** in combination with FIG. **16** shows a schematic block diagram of a DC-DC converter system **1500** in accordance with an eighth implementation of this description. More particularly, FIG. **15** shows a fixed current mode buck DC-DC converter system with valley current control topology. Similar to the timer unit **1200** of FIG. **12**, the timer unit **1200** is configured to generate a clock signal CLK with a cycle determined based on a larger one between the target switching cycle of the DC-DC converter system **1500** and an adjusted cycle determined based on the input and output voltages VIN and VOUT and a predetermined duration substantially equal to greater than the minimum on time T_{off_min} of the first switch **1502** of the DC-DC converter system **1500**. The control logic **1524** is configured to switch off the first switch **1502** responsive to, for example, a rising edge, of the clock signal CLK.

Table 4 lists the operation ranges of fixed frequency current mode DC-DC converter systems with different topologies in accordance with the fifth to eighth implementations of this description.

TABLE 4

Operation ranges of fixed frequency current mode DC-DC converter systems in accordance with implementation of this description				
Buck/ Boost	Topology	Ton and Toff	On Duty Cycle Range	VOUT/VIN range
Boost	Fixed frequency + Peak Current	Current Regulated, $T_{on} \geq T_{on_min}$ $T_{new} = \max\{T, T_{on_min} * VOUT / (VOUT - VIN)\}$	$(0,$ $1 - T_{off_min}/T)$	$(1, T / T_{off_min})$
	Fixed frequency + Valley Current	Current Regulated, $T_{off} \geq T_{off_min}$ $T_{new} = \max\{T, T_{off} * VOUT / VIN\}$	$(T_{on_min}/T,$ $1)$	$(T / (T - T_{on_min}),$ $\infty)$
Buck	Fixed frequency + Peak Current	Current Regulated, $T_{on} \geq T_{on_min}$ $T_{new} = \max\{T, T_{on_min} * VIN / VOUT\}$	$(0,$ $1 - T_{off_min}/T)$	$(0,$ $1 - T_{off_min}/T)$
	Fixed frequency + Valley Current	Current Regulated, $T_{off} \geq T_{off_min}$ $T_{new} = \max\{T, T_{off_min} * VIN / (VIN - VOUT)\}$	$(T_{on_min}/T,$ $1)$	$(T_{on_min}/T, 1)$

15

Referring to FIG. 17, a flow chart of a method 1700 for regulating a DC-DC converter system in accordance with an implementation of this description is shown. With reference to the DC-DC converter system 100 of FIG. 1, the DC-DC converter system includes the first switch 102 coupled between the switch node SW and a voltage supply node, for example, a ground node GND. The inductor 106 is coupled between the switch node SW and the voltage input node VIN, and second switch 104 is coupled between the switch node SW and the voltage output node VOUT. The first switch 102 is configured to periodically allow the inductor current IL to flow there through. Other topologies of current mode DC-DC converter systems with the same mechanism to sensing a load current and regulate the DC-DC converter system are possible as well, such as the DC-DC converter systems 300 to 1500 respectively shown in FIGS. 3-15.

Starting at step 1702, the control logic 124 generates a PWM signal to switch the first switch 102 from a first state, e.g. on state, to a second state, e.g. off state, through the driver unit 112.

At step 1704, the first timer 202 of the timer unit 122 starts timing a first duration T1, and substantially simultaneously, the second timer 212 of the timer unit 122 starts timing a second duration T2 that is determined based on input and output voltages VIN and VOUT of the DC-DC converter system 100 and a minimum duration of the first state of the first switch 102, i.e., the minimum on-time Ton_min, of the first switch 102.

In one example, for adaptive off-time/on-time current mode DC-DC converter systems such as the DC-DC converter systems 100 to 700 of FIGS. 1, 3, 5 and 7 with corresponding timer units 200 to 800 of FIGS. 2A, 4, 6 and 8, the first and second timers 202 and 212 are configured to start timing responsive to the first switch 102 being switched from the first state to the second state, the first duration T1 is a nominal duration of the second state of the first switch 102 determined based on a target switching frequency fsw of the DC-DC converter system and the input and output voltages, and the second duration T2 is an adjusted duration of the second state of the first switch determined based on the input and output voltages VIN and VOUT and a predetermined duration substantially equal to greater than a minimum duration of the first state of the first switch.

In the example with reference to the DC-DC converter system 100 of FIG. 1, the first duration T1 is a nominal duration of the off state, i.e., a nominal off time Toff, of the first switch 102, and provided in accordance with the equations below:

$$T1 = T \cdot VIN / VOUT \quad (11)$$

where T is a target switching cycle $T = 1/fsw$ of the DC-DC converter system 100, fsw is a target switching frequency of the DC-DC converter system 100.

The second duration T2 is an adjusted duration of the off state, i.e., an adjusted off-time, of the first switch 102 determined based on the input and output voltages VIN and VOUT and a predetermined duration Ton_min' substantially equal to or greater than a minimum duration of the on state, i.e., a minimum on-time Ton_min, of the first switch 102. The second duration T2 is provided in accordance with the equations below:

$$T2 = Ton_min' \cdot \frac{VOUT - VIN}{VOUT} \quad (12)$$

16

where Ton_min' is the predetermined duration substantially equal to or slightly greater, e.g. 10 ns greater, than the minimum on-time Ton_min of the first switch 902.

In another example, for fixed frequency current mode DC-DC converter systems such as the DC-DC converter systems 900 of FIG. 9 with a timer unit 1000 of FIG. 10, the first and second timers start timing responsive to the first switch 902 being switched from the second state to the first state, e.g., for the DC-DC converter system 900, from the off state to the on state. The first duration T1 is the target switching cycle T of the DC-DC converter system 900. The second duration T2 is an adjusted switching cycle determined based on the input and output voltages VIN and VOUT and a predetermined duration Ton_min' substantially equal to or greater than the minimum on time Ton_min of the first switch 902. The second duration T2 is provided in accordance with the equations below:

$$T2 = Ton_min' \cdot \frac{VOUT - VIN}{VOUT} \quad (13)$$

where Ton_min' is the predetermined duration substantially equal to or slightly greater, e.g. 10 ns greater, than the minimum on-time Ton_min of the first switch 902.

At step 1706, the sensing unit 114 generates a control signal Sc to switch the first switch 102 from the first state, e.g. the on state, to the second state, e.g. the off state, based on a difference between a sensed voltage Vs proportional to the inductor current IL and a difference between a reference voltage VREF and a feedback voltage VFB proportional to the output voltage VOUT.

In one example, for DC-DC converter systems with a peak current control topology, such as the DC-DC converter systems 100, 700, 900 and 1300 respectively shown in FIGS. 1, 7, 9 and 13, switching the first switch from the first state to the second state comprises switching the first switch from the on state to the off state responsive to a current through the first switch increasing to a peak value determined based on the difference between the feedback voltage VFB of the output voltage VOUT and the reference voltage VREF and expiration of a minimum on-time of the first switch.

In another example, for DC-DC converter systems with a valley current control topology, such as the DC-DC converter systems 300, 500, 1100 and 1500 respectively shown in FIGS. 3, 5, 11 and 15, switching the first switch from the first state to the second state comprises switching the first switch from an off state to an on state responsive to a current through the second switch decreasing to a valley value determined based on a difference between the feedback voltage VFB of the output voltage VOUT and a reference voltage VREF and expiration of a minimum off-time of the first switch.

At step 1708, the timer unit 122 generates an expiration signal S_T to switch the first switch 102 from the second state to the first state responsive to expiration of both the first and second timers.

The term "couple" is used throughout the specification. The term may cover connections, communications, or signal paths that enable a functional relationship consistent with the description of this description. For example, if device A generates a signal to control device B to perform an action, in a first example device A is coupled to device B by direct connection, or in a second example device A is coupled to device B through intervening component C if intervening

17

component C does not alter the functional relationship between device A and device B such that device B is controlled by device A via the control signal generated by device A.

Modifications are possible in the described embodiments, and other embodiments are possible, within the scope of the claims.

What is claimed is:

1. A DC-DC converter system, comprising:
 - a first switch coupled to a switching terminal of the DC-DC converter system, and configured to operate between first and second states;
 - a first timer comprising:
 - a first capacitive element with a first capacitance;
 - a first timer switch coupled in parallel with the first capacitive element;
 - a first current source coupled in series with the first capacitive element and configured to source or sink a first current to or from the first capacitive element; and
 - a first comparator with a first input terminal coupled to the first capacitive element, a second input terminal configured to receive a first reference voltage, and an output terminal configured to generate a first timer expired signal responsive to expiration of a first duration determined in response to the first capacitance, the first current source and the first reference voltage;
 - a second timer comprising:
 - a second capacitive element with a second capacitance;
 - a second timer switch coupled in parallel with the second capacitive element;
 - a second current source coupled in series with the second capacitive element and configured to source or sink a second current to or from the second capacitive element; and
 - a second comparator with a first input terminal coupled to the second capacitive element, a second input terminal configured to receive a second reference voltage, and an output terminal configured to generate a second timer expired signal responsive to expiration of a second duration determined in response to the second capacitance, the second current source and the second reference voltage;
 - a logic gate coupled to outputs of the first and second comparators, and configured to generate an expiration signal in response to expiration of both the first and second timers; and
 - a control logic unit, coupled between the logic gate and a control terminal of the first switch, configured to switch the first switch from the second state to the first state in response to the expiration signal, wherein:
 - the first timing switch is coupled to the control logic unit, and is configured to be switched off in response to the first switch being switched from the first state to the second state; and
 - the second timer switch is coupled to the control logic unit, and is configured to be switched off with the first timing switch.
2. The DC-DC converter system of claim 1, wherein:
 - the first comparator is configured to generate the first timer expired signal responsive to a voltage difference across the first capacitive element changing by a first voltage difference, wherein the first duration is deter-

18

mined by a target switch cycle and input and output voltages of the DC-DC converter system,

and

the second comparator is configured to generate the second timer expired signal responsive to a voltage difference across the second capacitive element changing by a second voltage difference, wherein the second duration is determined by a predetermined duration equal to or greater than a minimum duration of the first state of the first switch and the input and output voltages of the DC-DC converter system.

3. The DC-DC converter system of claim 2, wherein:

- a product of the first capacitance and a ratio of the first voltage difference to the first current equals to a duration of the second state of the first switch determined by the target switch cycle and input and output voltages of the DC-DC converter system, and
- a product of the second capacitance and a ratio of the second voltage difference to the second current equals to an adjusted duration of the second state of the first switch determined by the predetermined duration and the input and output voltages of the DC-DC converter system.

4. The DC-DC converter system of claim 1, further comprising:

- a one-shot signal generator configured to generate a one-shot signal in response to the first switch being switched from the second state to the first state, wherein the first timing switch includes a control terminal coupled to receive the one-shot signal, and is configured to be switched off when the one-shot signal is de-asserted,
- the first comparator is configured to generate the first timer expired signal responsive to a voltage difference across the first capacitive element changing by a first voltage difference, wherein the first duration is determined by a target switch cycle of the DC-DC converter system,

the second timing switch includes a control terminal coupled to receive the one-shot signal, and is configured to be switched off substantially simultaneously with the first timing switch, and

the second comparator is configured to generate the second timer expired signal responsive to a voltage difference across the second capacitive element changing by a second voltage difference, wherein the second duration is determined by a predetermined duration substantially equal to or longer than a minimum duration of the first state of the first switch and input and output voltages of the DC-DC converter system.

5. The DC-DC converter system of claim 4, wherein:

- a product of the first capacitance and a ratio of the first voltage difference to the first current equals to a target switch cycle of the DC-DC converter system, and
- a product of the second capacitance and a ratio of the second voltage difference to the second current equals to an adjusted switch cycle of the DC-DC converter system determined by the predetermined duration and the input and output voltages of the DC-DC converter system.

6. A system comprising:

- a first timer configured to generate a first timer expired signal in response to a first duration time;
- a second timer configured to generate a second timer expired signal in response to a second duration time;
- a logic circuit configured to generate an expiration signal in response to the first timer expired signal and the second timer expired signal; and

19

a control circuit configured to switch a switch from a second state to a first state in response to the expiration signal, wherein:
 the first timer is configured to start timing of the first duration time with the second timer starting timing of the second duration time in response to the switch being switched from the first state to the second state; and
 the system is capable of a target switching frequency determined by a capacitance and a resistance of a charging path in the first timer, and the first duration time is based on the target switching frequency of the system and an input voltage and an output voltage.

7. The system of claim 6, wherein:
 the second duration time is determined by a minimum off time of a second switch, the input voltage, and the output voltage.

8. The system of claim 6, wherein:
 in response to the first timer expired signal and the second timer expired signal being asserted, the expiration signal is generated by the logic circuit.

9. The system of claim 6, wherein:
 the logic circuit is an AND gate.

10. The system of claim 6, wherein:
 the first timer includes a first timer switch coupled in parallel with a first capacitor; and
 the second timer includes a second timer switch coupled in parallel with a second capacitor.

11. The system of claim 10, wherein:
 the first timer includes a first comparator with a first input terminal of the first timer coupled to the first capacitor, a second input terminal coupled to a first reference voltage; and

20

the second timer includes a second comparator with a first input terminal of the second timer coupled to the second capacitor, a second input terminal coupled to a second reference voltage.

12. The system of claim 11, wherein:
 the first input terminal of the first timer is coupled to a first current source; and
 the first input terminal of the second timer is coupled to a second current source.

13. The system of claim 12, wherein:
 the first current source is generated by an input voltage to the system; and
 the second current source is generated by an output voltage of the system.

14. The system of claim 6, wherein:
 the control circuit is configured to receive a control signal; and
 the control signal is determined by a current in the switch.

15. The system of claim 14, wherein:
 the control signal is asserted in response to a sensing voltage associated with the switch reaches a control voltage threshold.

16. The system of claim 14, wherein:
 the control circuit is configured to turn off the switch in response to the control signal.

17. The system of claim 6, wherein:
 the control circuit is coupled to a driver circuit; and
 the driver circuit is coupled to a control terminal of the switch.

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